Lab 3: Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk System

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1. Architectural Style Analysis

Style	Pros	Cons
Layered	Clear separation of concerns; easy to understand and build for a single self-contained device	Some inter-layer overhead; layers can become tightly coupled if not designed carefully
Microservices	Independent scaling and deployment; fault isolation; technology diversity	High operational/network complexity - unnecessary for one kiosk with fixed hardware
Client-Server	Centralized control; simple to keep menu/pricing data consistent	Single point of failure; scalability bottleneck if many kiosks depend on one server

2. Architecture Selection & Justification

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

The kiosk is organised into three layers: a Presentation layer (Touch Screen UI), a Business-logic layer (Order Manager, coordinating Payment Service and Receipt Printer), and a Data layer (Menu Catalog).

Reason 1 – Matches the scenario's scope: a single kiosk with fixed hardware (one touch screen, one printer, one card reader) has no need for independently scaled services; a layered structure gives structured modularity without microservices' network/deployment overhead.

Reason 2 – Simplicity and maintainability: clear separation between UI, order/payment logic and menu data makes the system easy to build within a fixed lab timeframe and easy to modify (e.g. changing menu prices) without touching UI or payment code.

Security advantage: only the Payment Service layer communicates with the card reader and handles cardholder data; the Presentation layer never touches raw payment data, limiting the attack surface and keeping sensitive processing isolated in one auditable layer.

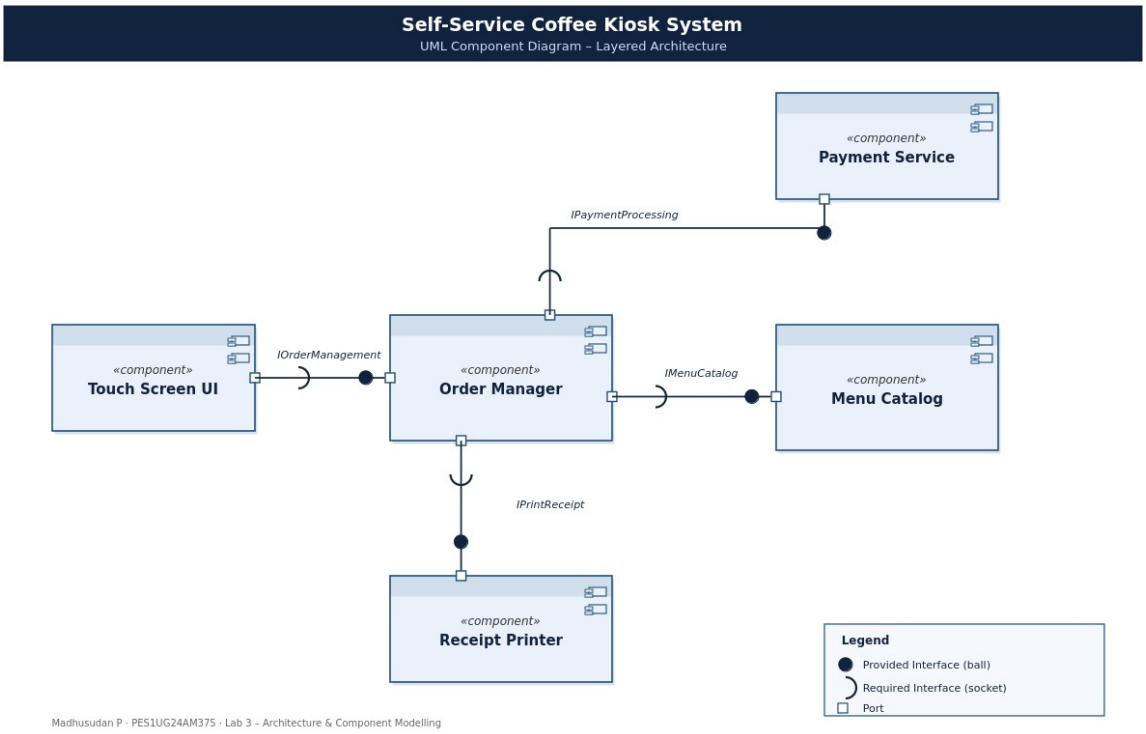
Performance benefit: because all layers run on the same local kiosk device, calls between layers are in-process rather than over a network (as microservices would require), giving the low-latency response the touch-screen interface needs.

3. Component Identification

Component	Responsibility
Order Manager (given)	Coordinates the end-to-end order workflow
Payment Service (given)	Handles credit-card payment processing
Touch Screen UI	Captures customer selections (drink type, size) and displays order status

Menu Catalog	Stores and serves coffee types, sizes and pricing
Receipt Printer	Drives the receipt printer hardware to print order details

4. UML Component Diagram



Ball notation marks a provided interface and socket notation marks a required interface; each required socket connects to the matching provided ball via an assembly connector. Small squares mark ports – the interaction points where each interface attaches to a component boundary.

5. Interfaces

Interface	Required by	Provided by	Purpose
IOrderManagement	Touch Screen UI	Order Manager	Sends selected items/size to order logic
IPaymentProcessing	Order Manager	Payment Service	Sends order total for card payment
IMenuCatalog	Order Manager	Menu Catalog	Fetches item names, sizes, prices
IPrintReceipt	Order Manager	Receipt Printer	Sends finalised order for printing